STOCHASTIC MODELING

MODULE 2 | LESSON 2

CHARACTERISTIC FUNCTIONS

Reading Time	30 minutes
Prior Knowledge	Probability Density Function (PDF)
Keywords	Characteristic Function (CF)

As we have already seen, one important input in Fourier-type option pricing methods is the characteristic function (CF) of the underlying stock price process. Here we dig a little bit deeper into the definition of CF.

1. Characteristic Function (CF)

The CF will come in handy because, for complex methods, directly observing the PDF is impossible and/or too computationally costly.

In this lesson, we will go over the mathematical foundations of CF and see a step-by-step example on how to derive it for a simple (Black-Scholes type) stock price process.

The **characteristic function**, $\varphi_X(u)$, of a real valued random variable X depicts the **distribution function uniquely**. It is defined as:

$$ \varphi_X (u) = \mathbb{E}[e^{iuX}]\ , \ with \ i = \sqrt{-1} $$

Which we can onveniently rewrite as:

$$ \varphi_X (u) = \int_{-\infty}^{\infty} e^{iuX} \,dF(x) = \int_{-\infty}^{\infty} e^{iuX}f(x) \,dx $$

where $F(x)$ and $f(x)$ denote the CDF and PDF of X, respectively.

In [1]:
```
from IPython.display import VimeoVideo

# Bigger video
VimeoVideo("1116720231", h="3298dbabb7", width=700, height=450)
```

Out[1]:

2. Deriving CF: Black-Scholes example

Next, we will go over the process of deriving a specific characteristic function from scratch. We will do that for the **Black-Scholes** model. However, the process is common to other models.

Before doing that, let's define a **Lemma** that will help us later on in the derivation.

- **Lemma**. Define $y = \hat{\mu}+i u \hat{\sigma}^2$, then:

$$ ius_T - \frac{(s_T - \hat{\mu})^2}{2\hat{\sigma}^2} = \frac{-(s_T - y)^2 + 2\hat{\mu}\hat{\sigma}^2 iu - u^2 \hat{\sigma}^4}{2\hat{\sigma}^2} $$

Proof

$$ ius_T - \frac{(s_T - \hat{\mu})^2}{2\hat{\sigma}^2} = \frac{2\hat{\sigma}^2 ius_T - (s_T^2 -2s_T\hat{\mu}+ \hat{\mu}^2) }{2\hat{\sigma}^2} = \frac{-s_T^2 + 2s_T\hat{\mu} + 2s_T iu\hat{\sigma}^2 - \hat{\mu}^2}{2\hat{\sigma}^2} = \frac{-(s_T - y)^2 + 2\hat{\mu}\hat{\sigma}^2 iu - u^2 \hat{\sigma}^4}{2\hat{\sigma}^2} $$

2.1. The Black-Scholes CF

Now, we will work on deriving the CF of a log-normal process, as is the case with stock prices in Black-Scholes.

Let's start by working with a normally-distributed variable $s = Ln S$, with $\mathbb{Q}(s_T)$ its risk-neutral PDF:

$$ \varphi_{s_T} (u) = \mathbb{E}[e^{ius_T}] = \int_{-\infty}^{\infty} e^{ius_T} \mathbb{Q}(s_T) \,ds_T = \int_{-\infty}^{\infty} e^{ius_T} \frac{1}{\sigma \sqrt{2\pi T}} e^{\frac{-\{s_T-[s_0 + (\mu - \frac{\sigma^2}{2})T]\}^2}{2\sigma^2T}} \,ds_T = \frac{1}{\sigma \sqrt{2\pi T}} \int_{-\infty}^{\infty} e^{ius_T - \frac{\{s_T-[s_0 + (\mu - \frac{\sigma^2}{2})T]\}^2}{2\sigma^2T}} \,ds_T $$

Let $\hat{\mu}= \left[s_0 + \left(\mu - \frac{\sigma^2}{2}\right) T \right]$ and $\hat{\sigma}= \sigma \sqrt{T}$, then:

$$ \varphi_{s_T} (u) = \frac{1}{\sigma \sqrt{2\pi T}} \int_{-\infty}^{\infty} e^{ius_T - \frac{\{s_T-[s_0 + (\mu - \frac{\sigma^2}{2})T]\}^2}{2\sigma^2T}} \,ds_T = \frac{1}{\hat{\sigma} \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ius_T - \frac{(s_T - \hat{\mu})^2 }{2\hat{\sigma}^2}} \,ds_T $$

Now, the **Lemma** defined before comes in handy. Applying it:

$$ \varphi_{s_T} (u) = \frac{1}{\hat{\sigma} \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ius_T - \frac{(s_T - \hat{\mu})^2 }{2\hat{\sigma}^2}} \,ds_T = \frac{1}{\hat{\sigma} \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{-(s_T - y)^2 + 2\hat{\mu}\hat{\sigma}^2 iu - u^2 \hat{\sigma}^4}{2\hat{\sigma}^2}} \,ds_T = e^{\frac{2\hat{\mu}\hat{\sigma}^2iu-u^2\hat{\sigma}^4}{2\hat{\sigma}^2}}\frac{1}{\hat{\sigma} \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{-(s_T - y)^2}{2\hat{\sigma}^2}} \,ds_T = e^{\hat{\mu}iu - \frac{1}{2}u^2\hat{\sigma}^2} \frac{1}{\hat{\sigma} \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{s_T - y}{\sqrt{2\hat{\sigma}^2}} \right) ^2} \,ds_T $$

Next, consider $z = \frac{s_T-y}{\sqrt{2\hat{\sigma}^2}}$, and the closed upper and lower bounds of s_T, $[-\gamma, \gamma]$,

$$ z = \frac{s_T - (\hat{\mu}+ iu\hat{\sigma}^2)}{\sqrt{2\hat{\sigma}^2}} = \frac{s_T - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}} - \frac{iu\hat{\sigma}}{\sqrt{2}} \\ \varphi_{s_T} (u) = e^{\hat{\mu}iu - \frac{1}{2}u^2\hat{\sigma}^2} \frac{1}{\hat{\sigma} \sqrt{2\pi}} \int_{\frac{-\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}}-\frac{iu\hat{\sigma}}{\sqrt{2}}}^{\frac{\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}}-\frac{iu\hat{\sigma}}{\sqrt{2}}} e^{-z^2}\sqrt{2\hat{\sigma}^2} \,dz $$

At this point, **Cauchy's integral theorem** will help in solving the integral analtically.

- **Cauchy's integral theorem**. If a function $f(z)$ is analytic in a simply connected domain Ω, then for any simply closed contour C in Ω,

$$ \int_C f(z) \,dz = 0 $$

Let a_i, for $i=1,2,3,4$ denote the corners of the rectangular contour C:

$$ a_1 = \frac{\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}}, \ \ a_2 = \frac{-\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}}, \ \ a_3 = \frac{-\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}}-\frac{iu\hat{\sigma}}{\sqrt{2}}, \ \ a_4 = \frac{\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}}-\frac{iu\hat{\sigma}}{\sqrt{2}} $$

By Cauchy's theorem we know that,

$$ \oint_{C \in \Omega} \sqrt{2\hat{\sigma}^2} e^{-z^2} \,dz = 0 \Leftrightarrow \int_{a_1}^{a_2} \sqrt{2\hat{\sigma}^2} e^{-z^2} \,dz + \int_{a_2}^{a_3} \sqrt{2\hat{\sigma}^2} e^{-z^2} \,dz + \int_{a_3}^{a_4} \sqrt{2\hat{\sigma}^2} e^{-z^2} \,dz + \int_{a_1}^{a_4} \sqrt{2\hat{\sigma}^2} e^{-z^2} \,dz = 0 $$

We can now solve the previous integrals separately and check their behavior when $\gamma \rightarrow \infty$. Let's start with those that do not contain any imaginary parts in the bounds:

$$ \lim_{\gamma \rightarrow \infty} \int_{a_1}^{a_2} \sqrt{2\hat{\sigma}^2} e^{-z^2} \,dz = -\lim_{\gamma \rightarrow \infty} \sqrt{2\hat{\sigma}^2} \int_{a_2}^{a_1} e^{-z^2} \,dz = - \sqrt{2\pi \hat{\sigma}^2} $$

Since we know that $\int_{-\infty}^{\infty} e^{-z^2} \,dz = \sqrt{\pi}$. Next, let's check those integrals with imaginary components in only one of the bounds, for example that between a_4 and a_1:

$$ \lim_{\gamma \rightarrow \infty} \int_{a_4}^{a_1} \sqrt{2\hat{\sigma}^2} e^{-z^2} \,dz = \lim_{\gamma \rightarrow \infty} \sqrt{2\hat{\sigma}^2} \int_{a_4}^{a_1} e^{-z^2} \,dz = 0 $$

It is easy to check that these will be 0, since we know that, for example, $\int_{\infty-iz}^{\infty} e^{-z} \,dz = 0$.

Finally, to fulfill Cauchy's theorem we know that:

$$ \int_{a_3}^{a_4} \sqrt{2\hat{\sigma}^2} e^{-z^2} \,dz = \sqrt{2\pi \hat{\sigma}^2} $$

Plugging the previous result of the characteristic function expression we had from before:

$$ \varphi_{s_T} (u) = e^{\hat{\mu}iu - \frac{1}{2}u^2\hat{\sigma}^2} \frac{1}{\hat{\sigma} \sqrt{2\pi}} \int_{\frac{-\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}}-\frac{iu\hat{\sigma}}{\sqrt{2}}}^{\frac{\gamma - \hat{\mu}}{\sqrt{2\hat{\sigma}^2}}-\frac{iu\hat{\sigma}}{\sqrt{2}}} e^{-z^2}\sqrt{2\hat{\sigma}^2} \,dz = e^{\hat{\mu}iu - \frac{1}{2}u^2\hat{\sigma}^2} \frac{\sqrt{2\pi \hat{\sigma}^2}}{\hat{\sigma} \sqrt{2\pi}} = e^{\hat{\mu}iu - \frac{1}{2}u^2\hat{\sigma}^2} $$

Replacing back $\hat{\mu}= \left[s_0 + \left(\mu - \frac{\sigma^2}{2}\right) T \right]$ and $\hat{\sigma}= \sigma \sqrt{T}$, yields the **Black-Scholes CF**

$$ \varphi_{s_T} (u) = e^{\hat{\mu}iu - \frac{1}{2}u^2\hat{\sigma}^2} = e^{\left(s_0 + \left(\mu - \frac{\sigma^2}{2}\right) T\right)iu - \frac{1}{2}u^2 \sigma^2 T} $$

3. Conclusion

One more time this has, arguably, been a mathematical intense lesson. Again, please try to keep up with the pace of the course by focusing on the intuition behind the derivation, and the overall process. The idea is that you understand how to perform these derivation and the founctions of the actual code, which *does not come out of the blue*.

In the following lesson we will learn how to implement all these methods into code.

See you there!

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